

Exercise Sheet #4: Electric Field

Discussion: Nov 10, 2025

Exercise 1: Discrete Charge Distribution

Two point charges of $+3.0 \mu\text{C}$ each are located at the points $(x, y) = (0, 2.0 \text{ m})$ and $(x, y) = (0, -2.0 \text{ m})$. Two additional point charges, each with charge q , are located at $(x, y) = (4.0 \text{ m}, 2.0 \text{ m})$ and $(x, y) = (4.0 \text{ m}, -2.0 \text{ m})$. The electric field at the point $(x, y) = (0, 0)$ due to the four charges is $\vec{E} = (4.0 \times 10^3 \hat{x}) \text{ N/C}$. Determine the value of q .

Solution

The electric field at the origin due to a point charge Q at position (x, y) is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r},$$

where $\hat{r}$ is the unit vector pointing from the charge to the field point.

For the two charges at $(0, 2 \text{ m})$ and $(0, -2 \text{ m})$, the electric field components at the origin point along the y -axis. Since they are equal in magnitude and opposite in direction, they cancel each other:

$$E_y = 0 \quad (\text{net field from the two } 3.0 \mu\text{C} \text{ charges})$$

The distance to the origin of each q charge is

$$r = \sqrt{(4.0)^2 + (2.0)^2} \text{ m} = \sqrt{20} \text{ m},$$

and the electric field at the origin due to one q charge has magnitude

$$E_q = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}.$$

At the origin, the y -components of the fields from the charges at $(4 \text{ m}, 2 \text{ m})$ and $(4 \text{ m}, -2 \text{ m})$ cancel, while the x -components add.

Thus, the total field at the origin due to both q charges is

$$|E_x| = 2 \cdot E_q \cos \theta,$$

where $\cos \theta = \frac{x}{r} = \frac{4}{4.472} = 0.894$.

Given that the total field at the origin is

$$E_x = 4.0 \times 10^3 \text{ N/C},$$

we have

$$q = -\frac{E_x \cdot 4\pi\epsilon_0 \cdot r^2}{2 \cdot \cos \theta} = -\frac{4.0 \times 10^3 \cdot 2\pi \cdot 8.854 \times 10^{-12} \cdot 20}{0.894} \text{ C} = -5.0 \mu\text{C}.$$

Exercise 2: Gauss's Law

A solid, non-conducting sphere of radius R carries a total charge Q distributed uniformly throughout its volume. Using Gauss's law, determine the electric field magnitude $E(r)$ as a function of radial distance r from the centre for:

- a) $0 \leq r < R$ (inside the sphere),
- b) $r \geq R$ (outside the sphere).

Give the final expressions in terms of Q , R , r , and the permittivity of free space ϵ_0 .

Solution

Because the charge is uniformly distributed over a sphere, the problem is spherically symmetric. The electric field must therefore be radial and have the same magnitude at every point on a spherical surface concentric with the charged sphere. Choose a spherical Gaussian surface of radius r , concentric with the charged sphere.

Gauss's law states:

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

On the chosen spherical surface the field is radial and constant in magnitude, and $d\mathbf{A}$ is radial, so the left-hand side becomes

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = E(r) \cdot 4\pi r^2.$$

We now compute Q_{enc} for the two regions.

- a) Outside the sphere: $r \geq R$

For $r \geq R$ the Gaussian surface encloses the entire charge Q . Thus $Q_{\text{enc}} = Q$, and Gauss's law gives:

$$E(r) 4\pi r^2 = \frac{Q}{\epsilon_0} \Rightarrow E(r) = \frac{Q}{4\pi\epsilon_0 r^2}, \quad r \geq R.$$

This is identical to the electric field of a point charge Q placed at the centre.

- b) Inside the sphere: $0 \leq r < R$ First compute the volume charge density ρ . The sphere's total volume is $\frac{4}{3}\pi R^3$, so

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3} = \frac{3Q}{4\pi R^3}.$$

The charge enclosed by a Gaussian sphere of radius $r < R$ is the charge occupying a smaller sphere of radius r :

$$Q_{\text{enc}} = \rho \cdot \frac{4}{3}\pi r^3 = \frac{3Q}{4\pi R^3} \cdot \frac{4}{3}\pi r^3 = Q \frac{r^3}{R^3}.$$

Apply Gauss's law:

$$E(r) 4\pi r^2 = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{Q}{\epsilon_0} \frac{r^3}{R^3}.$$

Solve for $E(r)$:

$$E(r) = \frac{Q}{4\pi\epsilon_0} \cdot \frac{r}{R^3}, \quad 0 \leq r < R.$$

Thus inside the sphere the magnitude of the field increases linearly with r .

Combined (piecewise) result:

$$E(r) = \begin{cases} \frac{Q}{4\pi\epsilon_0} \frac{r}{R^3}, & 0 \leq r < R, \\ \frac{Q}{4\pi\epsilon_0 r^2}, & r \geq R. \end{cases}$$

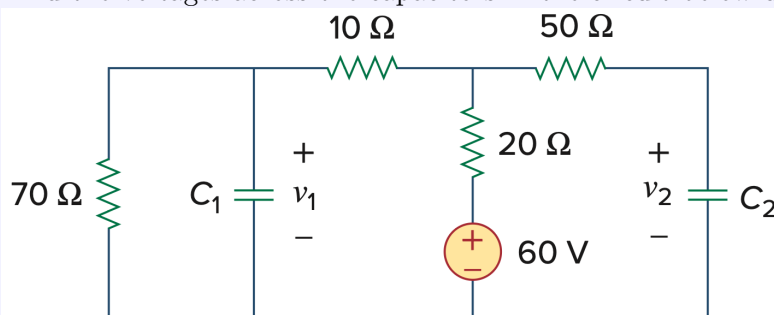
The direction is radial outward when $Q > 0$ and radial inward when $Q < 0$.

Remarks:

- Continuity at the surface: At $r = R$ the inside expression gives $Q/(4\pi\epsilon_0 R^2)$, which matches the outside expression evaluated at $r = R$. So the field is continuous at the surface.
- At the centre $r = 0$, $E(0) = 0$ (from the inside formula), consistent with symmetry.
- Outside the sphere the field is identical to that of a point charge Q at the centre.

Exercise 3: Capacitors in DC Steady State

Find the voltages across the capacitors in the circuit below under DC conditions.



Solution

Under DC steady-state conditions, the capacitors act as open circuits. Therefore, no current flows through C_1 or C_2 , and we can analyze only the resistive network.

Let the bottom node be the reference (0 V). The voltage at the top of the 60 V source is $V_C = 60$ V. Let V_A and V_B be the node voltages at the top of C_1 and the junction between the 10 Ω, 20 Ω, and 50 Ω resistors, respectively.

$$\begin{aligned} \text{KCL at node } A: \quad & \frac{V_A - 0}{70} + \frac{V_A - V_B}{10} + I_{C_1} = 0 \\ \Rightarrow & \frac{V_A - 0}{70} + \frac{V_A - V_B}{10} = 0 \end{aligned}$$

$$\begin{aligned} \text{KCL at node } B: \quad & \frac{V_B - V_A}{10} + \frac{V_B - 60}{20} + I_{C_2} = 0 \\ \Rightarrow & \frac{V_B - V_A}{10} + \frac{V_B - 60}{20} = 0 \end{aligned}$$

Multiply the first equation by 70:

$$V_A + 7(V_A - V_B) = 0 \quad \Rightarrow \quad 8V_A - 7V_B = 0$$

Multiply the second equation by 20:

$$2(V_B - V_A) + (V_B - 60) = 0 \implies -2V_A + 3V_B = 60$$

Substitute into each other:

$$-2\left(\frac{7}{8}V_B\right) + 3V_B = 60$$

$$\Rightarrow \left(-\frac{14}{8} + \frac{24}{8}\right)V_B = 60$$

$$\Rightarrow V_B = 60 \cdot \frac{8}{10} = 48 \text{ V}$$

$$V_A = \frac{7}{8}V_B = \frac{7}{8} \cdot 48 = 42 \text{ V} = v_1$$

Finally, let V_D be the trivial node (just one input and one output) voltage between the top of C_2 and the 50Ω resistor.

$$\text{KCL at node } D : \quad \frac{V_D - V_B}{50} + I_{C_2} = 0$$

$$I_{C_2} = 0 \text{ (DC steady state)} \Rightarrow \frac{V_D - V_B}{50} = 0 \Rightarrow V_D = V_B \Rightarrow v_2 = V_B$$

$$\boxed{v_1 = 42 \text{ V}, \quad v_2 = 48 \text{ V}}$$

Exercise 4: Plate Capacitor

We want to design a capacitor with capacitance $C = 1 \mu\text{F}$. Compute the length required for rectangular plates of 2 cm width if the dielectric is polyester ($\epsilon_r = 3.4$) of $15 \mu\text{m}$ thickness.

Solution

For a parallel-plate capacitor

$$C = \epsilon_0 \epsilon_r \frac{A}{d},$$

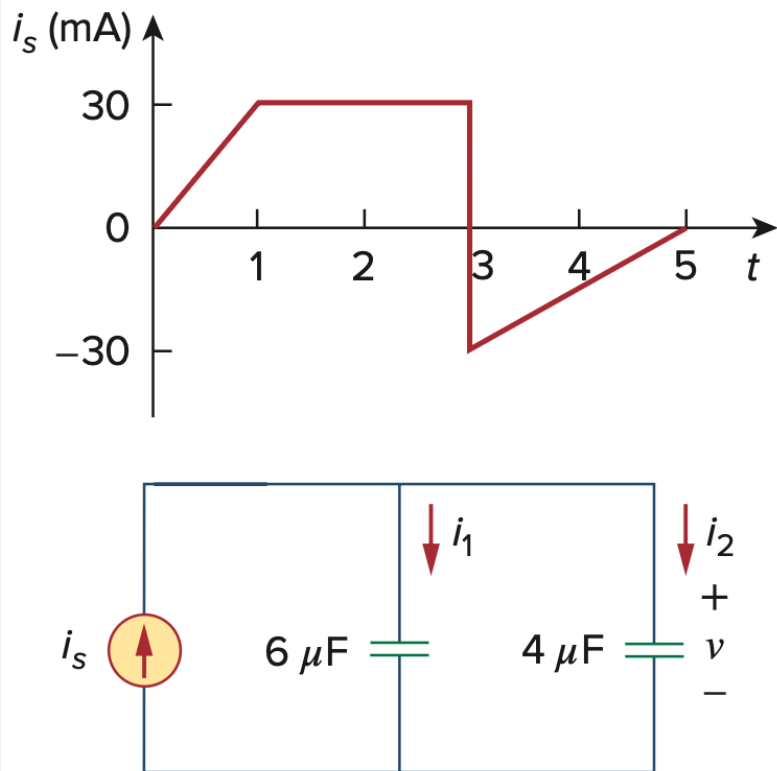
where A is the plate area and $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$ is the vacuum permittivity. For rectangular plates $A = wL$. Solving for L gives

$$L = \frac{C d}{\epsilon_0 \epsilon_r w} = \frac{1 \times 10^{-6} \cdot 15 \times 10^{-6}}{8.85 \times 10^{-12} \cdot 3.4 \cdot 0.02} \text{ m} = 24.93 \text{ m}.$$

Remarks: This large length is a consequence of the relatively small plate width and the modest relative permittivity; achieving $1 \mu\text{F}$ with a simple parallel-plate geometry and a thin polyester dielectric requires a very large plate area. Practical capacitors obtain large capacitances by using multiple layers, rolled foils, much higher-permittivity dielectrics, or much smaller effective separations (thin films, electrolytic constructions, etc.).

Exercise 5: Time-Dependent Capacitor Voltage and Current

If $v(0) = 0$, find $v(t)$, $i_1(t)$, and $i_2(t)$ in the circuit below.

**Solution**

$$i = C \frac{dv}{dt} \Rightarrow v = \frac{\int i \, dt}{C}$$

Here,

$$v(t) = \frac{\int i_s \, dt}{C_{eq}},$$

with the equivalent capacitance

$$C_{eq} = 6 \, \mu\text{F} + 4 \, \mu\text{F} = 10 \, \mu\text{F}.$$

For $0 < t < 1 \text{ s}$:

$$v(t) = \frac{\int 30t \, dt \, \text{mA} \cdot \text{s}}{10 \, \mu\text{F}} = \frac{15t^2 + \text{const}}{10} \, \text{kV}$$

$$v(0) = 0$$

$$\Rightarrow \text{const} = 0$$

$$\Rightarrow v(t) = 1.5t^2 \, \text{kV}$$

For $1 < t < 3 \text{ s}$:

$$v(t) = \frac{\int 30 \, dt}{10} \, \text{kV} = \frac{30t + \text{const}}{10} \, \text{kV}$$

$$v(1 \text{ s}) = 1.5 \cdot 1^2 \, \text{kV} = 1.5 \, \text{kV}$$

$$\Rightarrow \text{const} = 1.5 \cdot 10 - 30 \cdot 1 = -15$$

$$\Rightarrow v(t) = (3t - 1.5) \text{ kV}$$

For $3 < t < 5$ s:

$$v(t) = \frac{\int (\frac{30}{2}t - 75) dt}{10} \text{ kV} = \frac{7.5t^2 - 75t + \text{const}}{10} \text{ kV}$$

$$v(3\text{ s}) = [1.5t^2]_0^1 \text{ kV} + [3t - 1.5]_1^3 \text{ kV} = (1.5 - 0) \text{ kV} + (7.5 - 1.5) \text{ kV} = 7.5 \text{ kV}$$

$$\Rightarrow \text{const} = 7.5 \cdot 10 - 7.5 \cdot 9 + 75 \cdot 3 = 232.5$$

$$\Rightarrow v(t) = (0.75t^2 - 7.5t + 23.25) \text{ kV}$$

The currents through each capacitor are:

$$i_1 = C_1 \frac{dv}{dt}; \quad i_2 = C_2 \frac{dv}{dt}$$

For $0 < t < 1$ s:

$$i_1 = 6 \mu\text{F} \cdot 3t \text{ kV/s} = 18t \text{ mA}$$

$$i_2 = 4 \mu\text{F} \cdot 3t \text{ kV/s} = 12t \text{ mA}$$

For $1 < t < 3$ s:

$$i_1 = 6 \mu\text{F} \cdot 3 \text{ kV/s} = 18 \text{ mA}$$

$$i_2 = 4 \mu\text{F} \cdot 3 \text{ kV/s} = 12 \text{ mA}$$

For $3 < t < 5$ s:

$$i_1 = 6 \mu\text{F} \cdot (1.5t - 7.5) \text{ kV/s} = (9t - 45) \text{ mA}$$

$$i_2 = 4 \mu\text{F} \cdot (1.5t - 7.5) \text{ kV/s} = (6t - 30) \text{ mA}$$

Exercise 6: Energy of Capacitors

Two capacitors ($25 \mu\text{F}$ and $75 \mu\text{F}$) are connected to a 100 V source. Find the energy stored in each capacitor if they are connected in:

- a) parallel
- b) series

Solution

- a) The equivalent capacitance for parallel connection is

$$C_{eq} = C_1 + C_2 = 25 \mu\text{F} + 75 \mu\text{F} = 100 \mu\text{F}.$$

The energy stored in each capacitor is

$$E_1 = \frac{1}{2} \cdot C_1 \cdot V^2 = \frac{1}{2} \cdot 25 \times 10^{-6} \text{ F} \cdot (100 \text{ V})^2 = 125 \text{ mJ},$$

$$E_2 = \frac{1}{2} \cdot C_2 \cdot V^2 = \frac{1}{2} \cdot 75 \times 10^{-6} \text{ F} \cdot (100 \text{ V})^2 = 375 \text{ mJ}.$$

- b) The equivalent capacitance for series connection is

$$C_{eq} = \frac{C_1 \cdot C_2}{C_1 + C_2} = \frac{25 \mu\text{F} \cdot 75 \mu\text{F}}{25 \mu\text{F} + 75 \mu\text{F}} = 18.75 \mu\text{F}.$$

The voltage across each capacitor is

$$V_1 = \frac{Q}{C_1} = \frac{C_{eq} \cdot V}{C_1} = \frac{18.75 \mu\text{F} \cdot 100 \text{ V}}{25 \mu\text{F}} = 75 \text{ V},$$

$$V_2 = \frac{Q}{C_2} = \frac{C_{eq} \cdot V}{C_2} = \frac{18.75 \mu\text{F} \cdot 100 \text{ V}}{75 \mu\text{F}} = 25 \text{ V}.$$

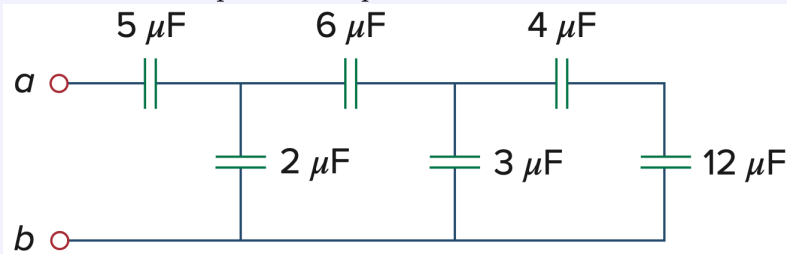
The energy stored in each capacitor is

$$E_1 = \frac{1}{2} \cdot C_1 \cdot V_1^2 = \frac{1}{2} \cdot 25 \times 10^{-6} \text{ F} \cdot (75 \text{ V})^2 = 70.31 \text{ mJ},$$

$$E_2 = \frac{1}{2} \cdot C_2 \cdot V_2^2 = \frac{1}{2} \cdot 75 \times 10^{-6} \text{ F} \cdot (25 \text{ V})^2 = 23.44 \text{ mJ}.$$

Exercise 7: Series and Parallel Capacitors

Determine the equivalent capacitance at terminals a-b of the circuit below.



Solution

$$C_{12,4,3} = \frac{12 \mu\text{F} \cdot 4 \mu\text{F}}{12 \mu\text{F} + 4 \mu\text{F}} + 3 \mu\text{F} = 6 \mu\text{F}$$

$$C_{12,4,3,6,2} = \frac{C_{4,12,3} \cdot 6 \mu\text{F}}{C_{4,12,3} + 6 \mu\text{F}} + 2 \mu\text{F} = 5 \mu\text{F}$$

$$C_{a-b} = \frac{C_{12,4,3,6,2} \cdot 5 \mu\text{F}}{C_{12,4,3,6,2} + 5 \mu\text{F}} = 2.5 \mu\text{F}$$